

Inductive vs. Deductive Reasoning: The Difference and How They Affect Your Well-Being

Inductive and deductive reasoning are thought patterns your brains uses to help you make decisions

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Person holding magnifying glass up to eye, with dotted paths, question mark and wheel cogs in background

Whether you realize it or not, you make about a gazillion decisions a day.

Do you need a coat today? What should you make for dinner? How should you respond to that text message? Does it even need a response?

Thankfully, our brains are problem-solving machines. They take in information, store it and use it later to help you make (mostly) reasonable decisions. Over and over and over again. All day. Every day.

Our brains come to decisions using several different forms of logic, including two of the most well-known types of reasoning: **inductive** and **deductive**. They're thought patterns that help us make sense of the world and take action accordingly.

But it's not always a perfect system. Because we humans aren't perfect beings (though the **perfectionists** among us may not like to admit it).

What's the difference between inductive vs. deductive reasoning? And what do they have to do with your well-being?

Health psychologist **Grace Tworek, PsyD**, explains the ways our reasoning can lead us to some less-than-reasonable conclusions.

Inductive vs. deductive reasoning

Logic and reasoning are core parts of the human experience. And how people make decisions and come to conclusions are complex and individualized processes.

Some experts describe a handful of different types of reasonings. Others theorize there are dozens of ways that people come to make decisions.

Two of the most basic forms of reasoning are inductive and deductive. And they can play a big part in constructing your worldview and influencing the choices you make.

What's the difference between inductive and deductive reasoning? At their core:

- **Inductive reasoning:** Allows you to reach a conclusion based on a specific observation.
- **Deductive reasoning:** Allows you to reach a conclusion based on a generalized premise.

We all use inductive and deductive reasoning all the time. There's nothing "wrong" with either. Neither is "better" than the other.

But when your reasoning is based on shaky foundations, your conclusions are more likely to lead you to less-than-fully-formed decisions. And that can impact your physical and mental well-being.

Let's look at some examples to better understand inductive and deductive reasoning.

How inductive reasoning works

Conclusion based on a specific observation.

"When you use inductive reasoning, you're essentially making an educated guess," Dr. Tworek shares. "You're using your past experiences and knowledge to predict what's likely to happen and take action based on that prediction."

Consider these examples:

- 1. Observation:** The weather app says it's going to be 20 degrees Fahrenheit (-7 degrees Celsius) today.
Reasoning: I've experienced that temperature before. I know that's cold.
Conclusion: I'll wear a coat today to keep comfortable.
- 2. Observation:** My phone bill is due today.
Reasoning: Last time I didn't pay the bill on time, I had to pay a late fee.
Conclusion: I'm going to pay the bill right now so I don't have to pay extra.
- 3. Observation:** I have an appointment tomorrow at 9 a.m. across town.
Reasoning: I've driven that way at that time before and the traffic is heavy.
Conclusion: I'll leave early so I won't be late.

In each example, there's a problem to be solved. You have experience with that problem. And inductive reasoning helped to use that experience to reach a well-thought-out solution.

Inductive reasoning downsides

But there can be a downside when your brain gets ... overzealous, so to speak, with inductive reasoning. That is, it can lead to you making decisions even when you don't have enough information to reach a fully reasonable conclusion.

For example:

- If you got bit by a dog, inductive reasoning could lead you to the conclusion that dogs are vicious and should be avoided. Probably an overreaction.
- If you had a lucky roll at the craps table, inductive reasoning could keep you throwing those dice longer. A recipe for big losses.
- If your kid and their friends go for an ill-advised helmet-free bike ride. No one gets hurt (thankfully). But they come to the conclusion that you're being overprotective when you insist they **wear proper head protection**. In reality, they don't know the dangers like you do because they're working from limited experience.

"Your worldview can become impacted when we follow theories based off of our own specific experiences," Dr. Tworek notes. "Someone who has the tendency to have positive takeaways from experiences may go into new experiences anticipating a positive outcome — a glass-half-full perspective. And vice versa. Negative experiences can lead to a mindset where poor outcomes are what's expected."

How deductive reasoning works

Conclusion based on a generalized premise.

When you use deductive reasoning, you start with a big-picture idea and then reason out a conclusion based on that framework.

“Deductive reasoning is almost like building with blocks,” Dr. Tworek illustrates. “You have the beginnings of a structure built with blocks — a premise. So, you rationalize when another block belongs within that structure.”

Consider these examples:

- 1. Observation:** I’m going to visit a friend in Seattle.
Premise: It rains a lot in Seattle.
Conclusion: It’s going to rain while I’m there. I’ll pack an umbrella.
- 2. Observation:** This ice cream has dairy.
Premise: Dairy upset my stomach.
Conclusion: This ice cream will give me a stomachache. I’m not going to eat it.
- 3. Observation:** I’m running low on clean underwear.
Premise: I need to do laundry to have clean clothes.
Conclusion: It’s time to do laundry.

Like inductive reasoning, deductive reasoning isn’t something you necessarily realize you’re doing. Because your brain is moving rapid-fire to assess the situation at hand and offer up a solution.

Deductive reasoning downsides

Relying on deductive reasoning in the wrong situations or applying a not-fully-formed premise to a situation at hand can lead to forming conclusions that don’t serve you well.

“Maybe you made a mistake on a challenging project at work and received negative feedback from your manager,” Dr. Tworek poses. “Now, you’re beginning a new project and feel challenged again. But the feedback from that other project has led you to believe that you don’t have the skills to complete this project, even if that’s not accurate. You conclude that you won’t do a good job on the task at hand, based on the (likely wrong) premise that you’re not good at your job.”

What to do about unhealthy reasoning

It’s only natural that you’ll use inductive and deductive reasoning (and other thought processes) day in and day out. Your brain is constantly in motion as you make decisions.

But the conclusions you come to don’t have to be on autopilot. Because, left unchecked, our reasoning isn’t always going to benefit us.

“There are times when our reasoning processes can make you feel limited, or it can perpetuate **negative self-talk** — meaning that you find you are not speaking kindly to yourself, or are even limiting yourself — as a result of the reasoning you are using,” Dr. Tworek shares.

Pausing to consider how you’re reacting to situations or why you’ve come to certain decisions can help you stop unhealthy thought patterns from taking over.

“Perhaps we were taught growing up that we can’t draw conclusions based on our own experiences, but that we must gather a large amount of information before coming to an answer. Or maybe we were raised in a way where one person’s experience created a set of rules or guidelines to follow,” she points out. “That can impact the types of reasoning that we default to and the situations that we use them in.”

What’s more, certain health conditions can affect your ability to reason logically. That includes ones like ADHD, brain tumors, mental health disorders and sleep disorders.

Recognizing unhealthy reasoning in yourself and reconsidering the premises that it's based on can be tough. After all, they're *your* thoughts. *Your* experiences. They live in *your* mind. So, it can be a challenge to step out of that construct. Regardless of how unhelpful that reasoning process actually is.

Engaging with a mental health professional can help. Because an outside perspective can help challenge unhealthy premises and help you find new strategies that can serve you better.

"A licensed mental health professional can assist in identifying thought patterns, understanding how we may be drawing certain conclusions or using certain methods of reasoning, and providing appropriate interventions," Dr. Tworek reassures.

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